MODULE 10

SLAM: Mapping While Finding Yourself

10.1 The chicken-and-egg problem

SIMPLE APPROACH

You are blindfolded in a new house and must draw a map of it. To draw a wall in the right place, you need to know where you are standing. But to know where you are standing, you need a map! SLAM is the clever trick of doing both a little at a time: take a step, feel a wall, guess where you are, add the wall to your map, take another step… and when you bump into a wall you already drew, you fix up your whole map.

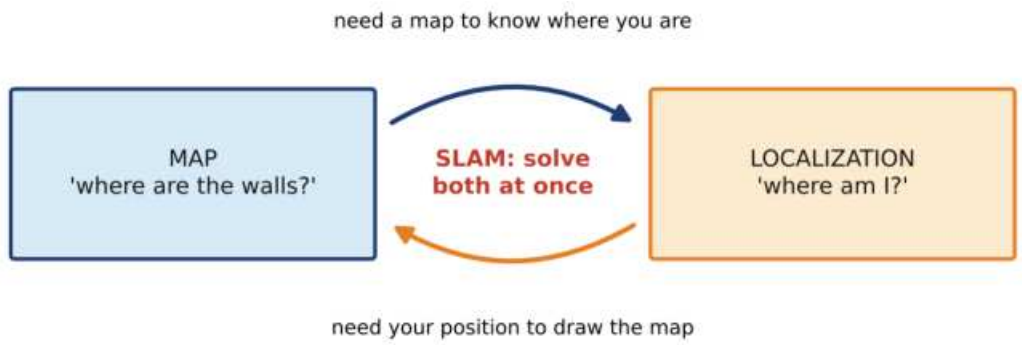


Figure 10.1: Simultaneous Localization And Mapping solves both questions together.

ENGINEER'S VIEW

SLAM estimates the trajectory $\mathbf{x}_{0:t}$ and the map $\mathbf{m}$ jointly from controls/odometry $\mathbf{u}_{1:t}$ and observations $\mathbf{z}_{1:t}$:

$$p(\mathbf{x}_{0:t}, \mathbf{m} \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}).$$

Surveyors know this problem as *a traverse run through new terrain while simultaneously locating the detail points*; each new shot both extends the traverse and adds map points, and the whole network is adjusted at the end. This module builds the classical tools — motion model, measurement model, Bayes filter, Kalman filters, particle filters — following Claus Brenner's well-known SLAM lecture series. Modules 11 and 12 add scan matching and the modern factor-graph form used by LIO-SAM.

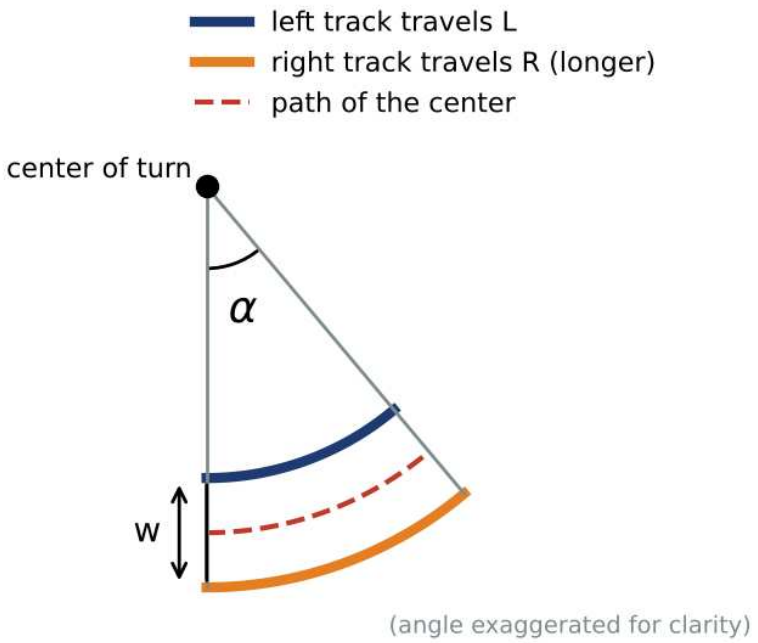


Figure 10.2: A two-track (differential drive) platform: when one side travels farther, the platform turns.

10.2 The motion model

KEY EQUATION — Differential-drive motion model

$$\alpha = \frac{R - L}{w}, \qquad r = \frac{L}{\alpha}, \qquad \begin{aligned} x' &= x + \left(r + \tfrac{w}{2}\right)\left(\sin(\theta + \alpha) - \sin\theta\right) \\ y' &= y - \left(r + \tfrac{w}{2}\right)\left(\cos(\theta + \alpha) - \cos\theta\right) \\ \theta' &= \theta + \alpha \end{aligned}$$

What every symbol means

Symbol	Meaning	Units
x, y, θ	pose before the move (position and heading)	m, m, rad
L, R	distance traveled by the left and right tracks	m
w	track width (distance between the tracks)	m
α	how much the platform turned	rad
r	turning radius of the *left* track; $r + w/2$ is the radius of the center	m
x', y', θ'	predicted pose after the move	m, m, rad

If $L = R$ the platform drives straight: $x' = x + L\cos\theta$, $y' = y + L\sin\theta$, $\theta' = \theta$.

WORKED EXAMPLE — predicting the next pose

Given: pose $(10.00,\ 5.00,\ 0°)$; $L = 0.40$ m, $R = 0.45$ m, $w = 0.30$ m.

Step 1 — turn angle

$\alpha = (0.45 - 0.40)/0.30 = 0.16667$ rad $= 9.549°$

Step 2 — radius

$r = 0.40/0.16667 = 2.400$ m; center radius $r + w/2 = 2.550$ m

Step 3 — x'
$10 + 2.550(\sin 9.549^\circ - \sin 0^\circ) = 10 + 2.550(0.16590) = 10.4230$ m

Step 4 — y'
$5 - 2.550(\cos 9.549^\circ - 1) = 5 - 2.550(-0.01386) = 5.0353$ m

Step 5 — θ'
$0 + 9.549^\circ = 9.549^\circ$

Result: new pose $(10.423,\ 5.035,\ 9.55^\circ)$: about 42 cm forward while turning 9.55° left.

On a mapping vehicle the same "prediction" role is played by IMU strapdown integration (Module 7) or wheel odometry.

10.3 Uncertainty grows when you move and shrinks when you look

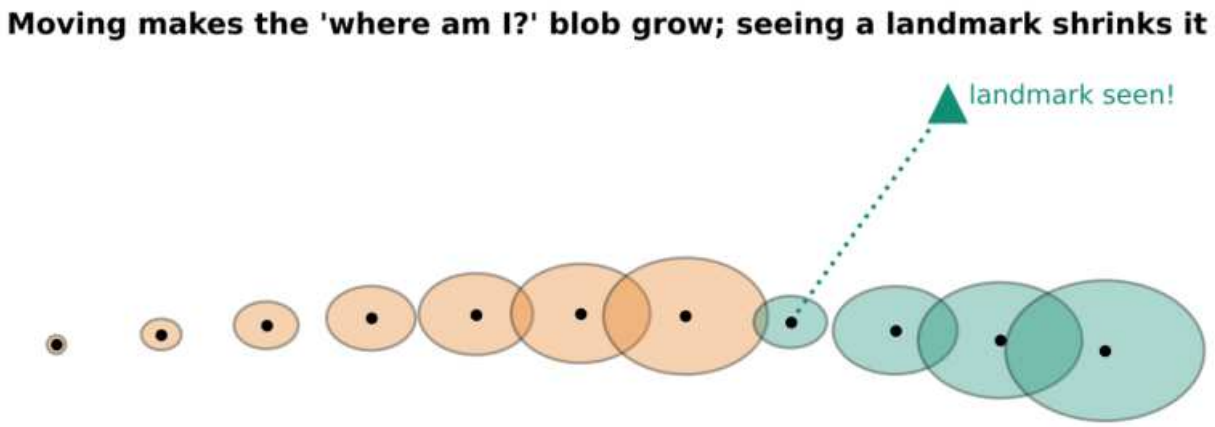


Figure 10.3: Each prediction inflates the 2σ uncertainty ellipse; one good landmark observation collapses it.

KEY EQUATION — The Bayes filter — predict, then correct

$$\underbrace{\overline{bel}(\mathbf{x}_t) = \int p(\mathbf{x}_t \mid \mathbf{u}_t, \mathbf{x}_{t-1})\, bel(\mathbf{x}_{t-1})\, d\mathbf{x}_{t-1}}_{\text{PREDICT (move)}} \qquad \underbrace{bel(\mathbf{x}_t) = \eta\, p(\mathbf{z}_t \mid \mathbf{x}_t)\, \overline{bel}(\mathbf{x}_t)}_{\text{CORRECT (measure)}}$$

What every symbol means

Symbol	Meaning	Units
$bel(\mathbf{x}_t)$	"belief": probability distribution of where we are at time t	—
$\overline{bel}$	predicted belief, before using the new measurement	—
$p(\mathbf{x}_t \mid \mathbf{u}_t, \mathbf{x}_{t-1})$	motion model with noise: where could I be after this control?	—
$p(\mathbf{z}_t \mid \mathbf{x}_t)$	measurement model (likelihood): how probable is this observation if I'm at $\mathbf{x}_t$?	—
η	normalizing constant so probabilities add to 1	—
$\int \ldots d\mathbf{x}_{t-1}$	add up over every place we might have started	—

10.4 The Kalman filter in one dimension

SIMPLE APPROACH

Two friends guess how tall a tree is. One is a careful measurer; the other is just eyeballing. You trust the careful one more, so your final answer is *between* their guesses but closer to the careful one. The Kalman filter does exactly that, with numbers that say *how much* closer.

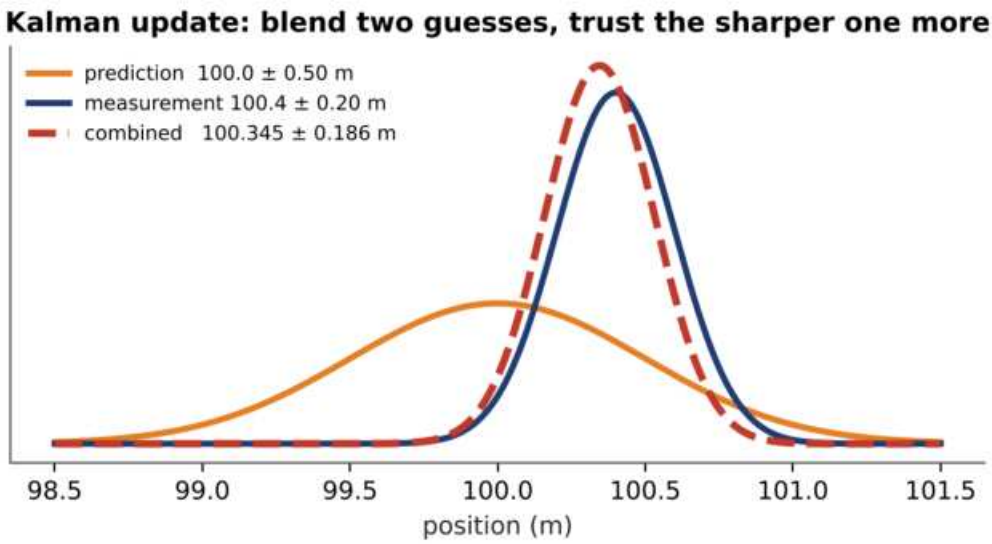


Figure 10.4: A wide prediction and a sharp measurement combine into an even sharper estimate.

KEY EQUATION — 1-D Kalman update

$$K = \frac{\bar{\sigma}^2}{\bar{\sigma}^2 + \sigma_z^2}, \qquad \mu = \bar{\mu} + K(z - \bar{\mu}), \qquad \sigma^2 = (1 - K)\,\bar{\sigma}^2$$

What every symbol means

Symbol	Meaning	Units
$\bar{\mu}, \bar{\sigma}^2$	predicted value and its variance	m, m^2
z, σ_z^2	measurement and its variance	m, m^2
K	Kalman gain, between 0 (ignore the measurement) and 1 (believe it completely)	—
$z - \bar{\mu}$	the *innovation*: how surprised we are	m
μ, σ^2	updated (posterior) value and variance — always smaller than both inputs' variances	m, m^2

WORKED EXAMPLE — 1-D Kalman update

Given: prediction $\bar{\mu} = 100.0$ m, $\bar{\sigma} = 0.50$ m; measurement $z = 100.4$ m, $\sigma_z = 0.20$ m.

Step 1 — variances

$\bar{\sigma}^2 = 0.25$ m^2; $\sigma_z^2 = 0.04$ m^2

Step 2 — gain

$K = 0.25/(0.25 + 0.04) = 0.8621$

Step 3 — update

$\mu = 100.0 + 0.8621 \times 0.4 = 100.345$ m

Step 4 — new variance

$\sigma^2 = (1 - 0.8621) \times 0.25 = 0.03448$ m$^2 \Rightarrow \sigma = 0.186$ m

Result: 100.345 ± 0.186 m. Note that 0.186 m beats even the measurement's 0.20 m.

ENGINEER'S VIEW

Surveyor's equivalence. The 1-D Kalman update is the weighted mean you already know. With weights $w = 1/\sigma^2$: $\mu = (w_1\bar{\mu} + w_2 z)/(w_1 + w_2) = (4 \times 100.0 + 25 \times 100.4)/29 = 100.345$ m, and $\sigma^2 = 1/(w_1 + w_2) = 1/29 = 0.0345$ m^2. Same answer.

10.5 The extended Kalman filter (EKF)

Real motion and measurement models contain sines and cosines — they are *non-linear*. The EKF linearizes them at the current estimate using **Jacobians** (tables of partial derivatives: "if I nudge this input, how much does each output change?").

KEY EQUATION — EKF prediction

$$\bar{\mathbf{x}} = g(\mathbf{x}, \mathbf{u}), \qquad \bar{\mathbf{P}} = \mathbf{G}\,\mathbf{P}\,\mathbf{G}^\top + \mathbf{Q}, \qquad \mathbf{G} = \frac{\partial g}{\partial \mathbf{x}}$$

What every symbol means

Symbol	Meaning	Units
g	the motion model (previous section)	—
$\mathbf{P}, \bar{\mathbf{P}}$	covariance matrix of the state before and after prediction; its diagonal holds the variances	m^2, rad^2
$\mathbf{G}$	state-transition Jacobian	—
$\mathbf{Q}$	process noise: extra uncertainty added by the motion itself	m^2, rad^2

WORKED EXAMPLE — EKF prediction for our motion

Given: the motion above; $\mathbf{P} = \mathrm{diag}(0.01, 0.01, 0.001)$, $\mathbf{Q} = \mathrm{diag}(0.002, 0.002, 0.0005)$.

Step 1 — Jacobian

Differentiate x' and y' by θ: $\partial x'/\partial\theta = 2.55(\cos 9.549^\circ - 1) = -0.0353$, $\partial y'/\partial\theta = 2.55 \sin 9.549^\circ = 0.4230$:

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & -0.0353 \\ 0 & 1 & 0.4230 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 2 — $\mathbf{G}\mathbf{P}\mathbf{G}^\top$

$\begin{pmatrix} 0.01000 & -0.00001 & -0.00004 \\ -0.00001 & 0.01018 & 0.00042 \\ -0.00004 & 0.00042 & 0.00100 \end{pmatrix}$ (heading uncertainty leaks into y)

Step 3 — add $\mathbf{Q}$

$\bar{\mathbf{P}} = \begin{pmatrix} 0.01200 & -0.00001 & -0.00004 \\ -0.00001 & 0.01218 & 0.00042 \\ -0.00004 & 0.00042 & 0.00150 \end{pmatrix}$

Result: 1σ: x 11.0 cm, y 11.0 cm, heading $\sqrt{0.0015} = 0.0387$ rad $= 2.22°$. Uncertainty grew, as expected.

10.5.1 Correcting with a landmark

KEY EQUATION — Range–bearing measurement model and EKF correction

$$\hat{r} = \sqrt{\Delta x^2 + \Delta y^2}, \quad \hat{\beta} = \operatorname{atan2}(\Delta y, \Delta x) - \theta; \qquad \mathbf{S} = \mathbf{H}\bar{\mathbf{P}}\mathbf{H}^{\mathsf{T}} + \mathbf{R}_z, \quad \mathbf{K} = \bar{\mathbf{P}}\mathbf{H}^{\mathsf{T}}\mathbf{S}^{-1}$$

$$\mathbf{x} = \bar{\mathbf{x}} + \mathbf{K}(\mathbf{z} - \hat{\mathbf{z}}), \qquad \mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\bar{\mathbf{P}}, \qquad \mathbf{H} = \begin{pmatrix} -\Delta x/\hat{r} & -\Delta y/\hat{r} & 0 \\ \Delta y/\hat{r}^2 & -\Delta x/\hat{r}^2 & -1 \end{pmatrix}$$

What every symbol means

Symbol	Meaning	Units
$\Delta x, \Delta y$	landmark position minus predicted vehicle position	m
$\hat{r}, \hat{\beta}$	predicted range and bearing to the landmark	m, rad
$\mathbf{z}$	actual measured range and bearing	m, rad
$\mathbf{H}$	measurement Jacobian	—
$\mathbf{R}_z$	measurement noise covariance	m^2, rad^2
$\mathbf{S}$	innovation covariance	—
$\mathbf{K}$	Kalman gain matrix (3×2)	—

WORKED EXAMPLE — EKF correction with a pole at (25, 8)

Given: predicted pose $(10.423,\ 5.035,\ 9.549°)$ and $\bar{\mathbf{P}}$ from above. Measured $r = 14.95$ m, $\beta = 2.10°$. $\mathbf{R}_z = \operatorname{diag}(0.05^2,\ (1°)^2)$.

Step 1 — predicted measurement

$\Delta x = 14.577$, $\Delta y = 2.965$; $\hat{r} = \sqrt{221.277} = 14.875$ m; $\hat{\beta} = 11.496° - 9.549° = 1.947°$

Step 2 — innovation

$\delta r = 14.95 - 14.875 = 0.075$ m; $\delta\beta = 2.10 - 1.947 = 0.153°$

Step 3 — Jacobian

$\mathbf{H} = \begin{pmatrix} -0.9799 & -0.1993 & 0 \\ 0.0134 & -0.0659 & -1 \end{pmatrix}$

Step 4 — gain

$\mathbf{S} = \begin{pmatrix} 0.01450 & 0.00005 \\ 0.00005 & 0.00192 \end{pmatrix}$, $\mathbf{K} = \begin{pmatrix} -0.811 & 0.124 \\ -0.164 & -0.635 \\ -0.0006 & -0.798 \end{pmatrix}$

Step 5 — update state

$\mathbf{x} = (10.423, 5.035, 9.549°) + \mathbf{K}(0.075,\ 0.00267\ \text{rad}) = (10.363,\ 5.021,\ 9.424°)$

Step 6 — update covariance

1σ: x 4.9 cm, y 10.5 cm, heading 0.96°

Result: the pole pulled the pose 6 cm back and tightened x from 11.0 to 4.9 cm. y barely improved because the pole is almost straight ahead: a range measurement only constrains the direction it points.

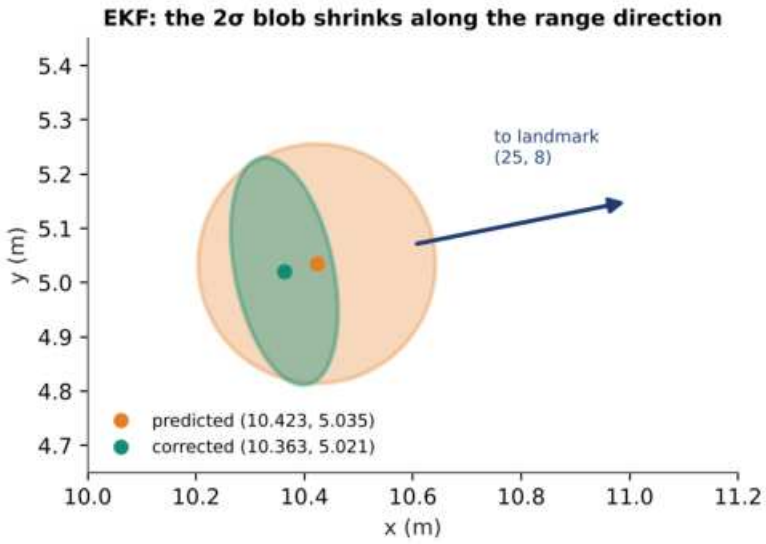


Figure 10.5: The 2σ ellipse before (orange) and after (teal) seeing the landmark.

10.6 Particle filters and EKF-SLAM

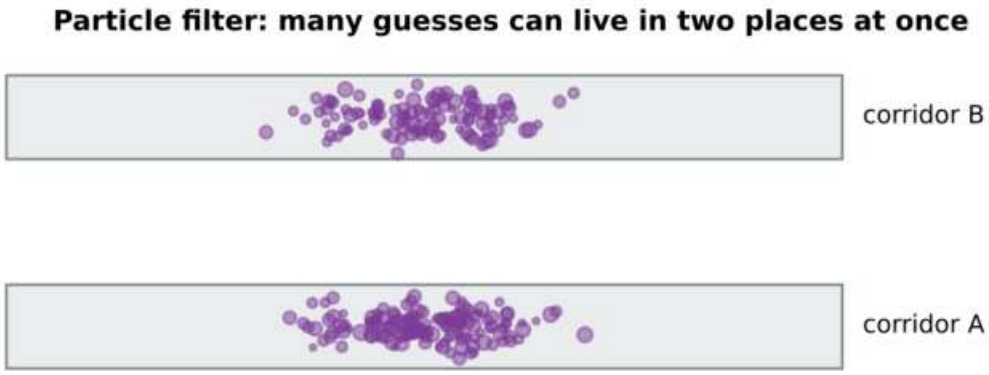


Figure 10.6: Particles can represent "I'm in corridor A *or* corridor B."

ENGINEER'S VIEW

Particle filters represent the belief by many weighted samples: move each with the noisy motion model, re-weight each by the measurement likelihood, then resample (clone heavy particles, drop light ones). They handle multi-modal situations a single Gaussian cannot. **FastSLAM** gives each particle its own small Kalman filter per landmark.

EKF-SLAM enlarges the state to hold the landmarks too: $\mathbf{x} = (x, y, \theta, m_{1x}, m_{1y}, \ldots)$. When a new landmark is seen, it is appended; when an old one is re-observed, the correlations update the pose *and every correlated landmark*. That joint estimation is the meaning of "simultaneous." Its cost grows with the square of the number of landmarks, which is why large-scale systems moved to sparse factor graphs (Module 12).

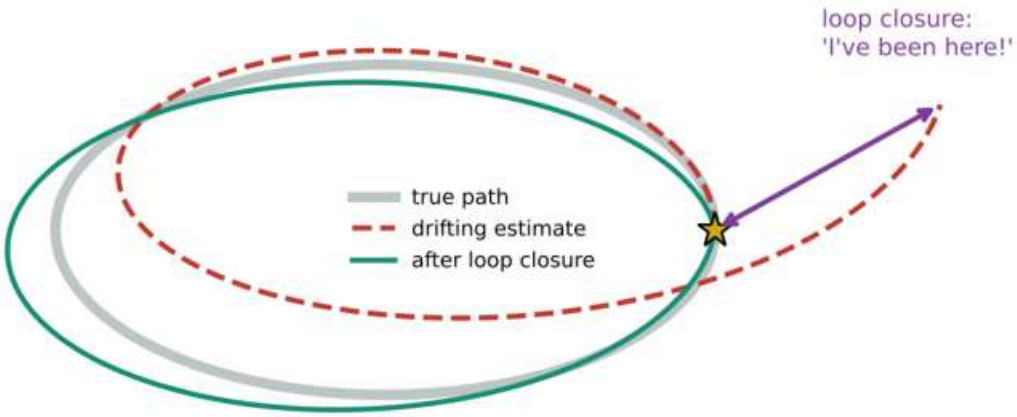


Figure 10.7: Odometry drifts like an open traverse; recognizing the start point closes the loop and distributes the misclosure.

10.7 Drift and loop closure

SURVEYOR'S FIELD TIP

Plan routes like traverses: return to the start, cross earlier paths, and add cross-ties in long corridors. Every revisit is a free closure. A route that never revisits anything is an open traverse — its end can be far off and nothing in the data will tell you.

TRY IT YOURSELF

10.1 Redo the motion-model example with $L = R = 0.40$ m.
10.2 Prediction 50.00 ± 0.30 m; measurement 50.20 ± 0.30 m. Find K, μ and σ.
10.3 (Simple) Why does seeing a landmark make the robot more sure of where it is?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 163.

SOURCES FOR THIS MODULE

Thrun, S., Burgard, W. & Fox, D. (2005), *Probabilistic Robotics*, MIT Press; Brenner, C., SLAM lecture series (Units A–G), Leibniz University Hannover.